Chandrashekhar Hiremath

Gen AI Engineer

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Dedicated Python developer with Overall 6 years and 4 years of experience in designing and implementing robust, scalable solutions. Proficient in building data-driven applications and optimizing performance for seamless functionality. Seeking to contribute to innovative projects and advance professional growth in a dynamic environment.

Experience

Infosys Pvt ltd May 2022 - present

Technical Diagnostic Assistance Using Generative AI

Overview: Dealer Technicians attend the complaints by their experience or by referring few documents. In the current context when Dealer Technicians search for the information in KGIS or KDP Global search they may not find the relevant information or there could be too many documents to refer. In cases where they are not able to find the right information they create a Techline case. GenAl seeks to reduce unnecessary case generation by offering immediate solutions to technicians during customer interactions. The proposed strategy involves displaying relevant information and document links in a pop-up window as technicians input data into the Techline Portal by using **Semantic search with RAG**

Techincal Detils:

- **Data Ingestion** process involves extracting data from SQL database, transforming it and loading it into both Cosmos DB and Blob Storage
- **Data Processing** handle pdf files, extract meaningful content, and create structured document for indexing.
- **Indexing :** documents are organized into different indexes based on their menu IDs.
- The **Ensemble Retriever** combines multiple retrievers allowing the system to query across different indexes.
- The retrived chunks data are concatenated into single sting to form the context text. This context text is then used to generate the responses.

Technologies Involved : Python, Langchain, SQL, Flask, Oracle DB, Azure cosmos DB, Blob Storage, Azure Open AI Model, RAG Solution.

Banking Project

Conducted a comprehensive analysis of the existing UNIX-based application to identify its functionalities. dependencies, and limitations.
Designed the architecture for the Python-based solution, considering scalability, modularity, and performance optimization.
Identified suitable Python libraries, frameworks, and tools to support the desired functionalities and system requirements.
Collaborated with the development team to define coding standards, best practices, and guidelines for the new Python codebase
Developed a step-by-step migration plan, ensuring a smooth transition from UNIX to Python.
Ported and refactored the existing UNIX codebase to Python, maintaining the core functionalities and adhering to the defined architecture.
Conducted thorough testing and debugging to identify and resolve any issues introduced during the migration process

Technologies Involved : Python, Unix , MySQL, Pycharm, PostgressSQL.

Networking Project:

Retrieved free IP addresses from Easy IP and updated the state to the " Assigned" for deployment.
Updated IP address records and related data in Netcracker, ensuring synchronization with network inventory. Automated request approval process, reducing lead time for network service deployments.
Updated customer and network configuration details in the Config Engine web portal.
Downloaded and manually reviewed configuration files for accuracy before uploading them to the portal.

Performed detailed manual checks of network configuration files to ensure compliance with customer specifications.
Created a RESTful API using Flask to run the automation process, allowing remote execution of tasks such as IP address assignment, request creation, and configuration updates.
Designed a web-based interface with HTML and CSS to allow users to interact with the API for managing IP addresses and configuration tasks.

Technologies Involved: Python, Selenium, Pywinauto, Flask, Pyautogui. HTML, CSS, Linux.

Cadmaxx Solutions Pvt ltd — Nov 2016 - Jun 2018

- InvolvedInCAEModellingto Build Car Body trims, Dash Board etc, to perfom CrashTest

LG Soft India Pvt ltd — July 2018 - May 2020

- Involved In CAE Modelling Activitiies for Washing Machine body Parts etc , tested Strengh analysis.

Otcalent services Pvt ltd — Jun 2020 - Sep 2021

Python Engineer Intern

- Write, test, and debug clean, maintainable, and efficient code as per project requirements.
- Work closely with senior developers, designers, and product managers to understand and implement software features.
- Identify and resolve bugs, performance issues, and bottlenecks in the application.
- Maintain proper documentation for code, processes, and APIs to ensure future maintainability.

Responsbilities

- Designed and implemented scalable Python solutions for various business requirements. Built and maintained automated test scripts using Selenium to validate web application functionality.
- Performed basic Unix/Linux system operations, including file handling, process monitoring, and shell scripting for task automation.
- Integrated front-end designs with Flask backend to deliver end-to-end web solutions. Explored and
- Implemented Lang Chain-based solutions to enhance automation and data workflows. Integrated
- AI/LLM capabilities for dynamic content generation and process optimization.
- Worked closely with cross-functional teams, including QA and DevOps, to ensure smooth development and deployment pipelines. Followed coding standards, performed code reviews, and documented processes to maintain code quality and knowledge sharing.

Education

High School — 2008

Visvesvaraya Technological University — 2016

- Bachelor of Engineering
- GPA: 60 %

Skills

- **Programming Languages:** Python
- **Web Frameworks:** Django, Flask, Fastapi.
- **Automation & Testing:** Selenium
- **Cloud Platforms:** Microsoft Azure (including Blob Storage, Azure Cosmos DB)
- **Databases:** SQL, Oracle DB, Azure Cosmos DB
- **AI & Machine Learning:** Generative AI, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Langgraph.
- **Tools & IDEs:** Git, Visual Studio Code (VS Code), PyCharm, Postman